Cosmic Beacons: Mapping Ammonia Masers in the W51 Star Forming Region

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Credit: NASA/SOFIA/Wanggi Lim, James De Buizer; NASA/JPL-Caltech

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High-Mass Star Forming Regions

- High Mass Stars are significant drivers of galactic evolution.
- Responsible for chemical and physical activity in the ISM.
- High volume of research, yet still poorly understood due to resolution limitations.

Hubble image of the S106 high mass star forming region in the Cygnus cloud. S106-IR is a newly formed star in the middle of the image. False coloring comes from using Blue for the Hydrogen and Red for the Dust

Credit: NASA & ESA

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W51

- The W51 region is an inner-galaxy target with an extremely high SFR.
- Recorded the second brightest H2O maser in the galaxy.
- One of the brightest NH3 masers in the galaxy.
- Targets for observation: W51-IRS2, W51-E.
- This talk focuses on W51-IRS2.

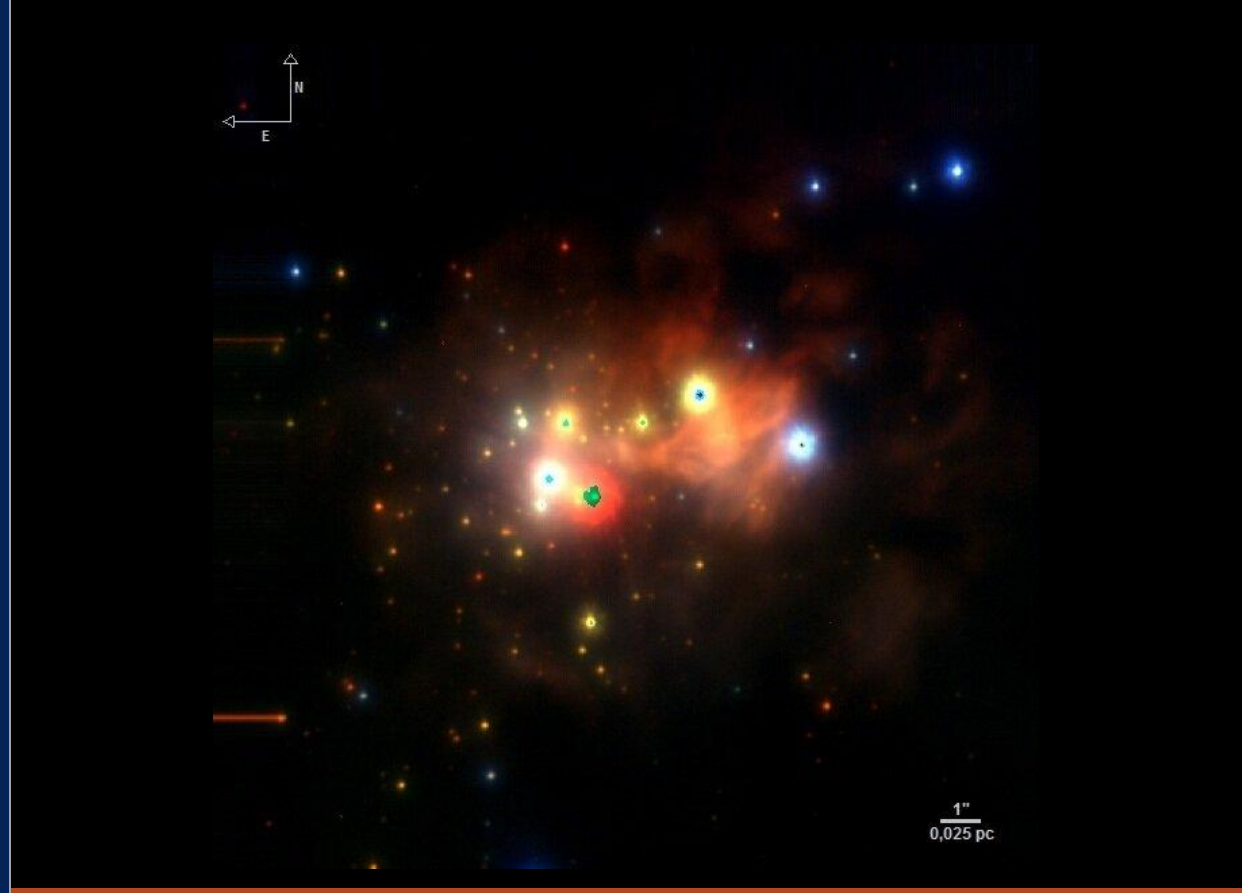

Image of W51 IRS2 in false colors. J corresponds to blue, H corresponds to green and K to red. It is possible to observe the reddishness of the studied region while the foreground objects present a predominance of blue. The complete image covers an area of 22 × 22 arc seconds.

Martins, Evelyn & Barbosa, C. (2014). População estelar da W51 IRS2.

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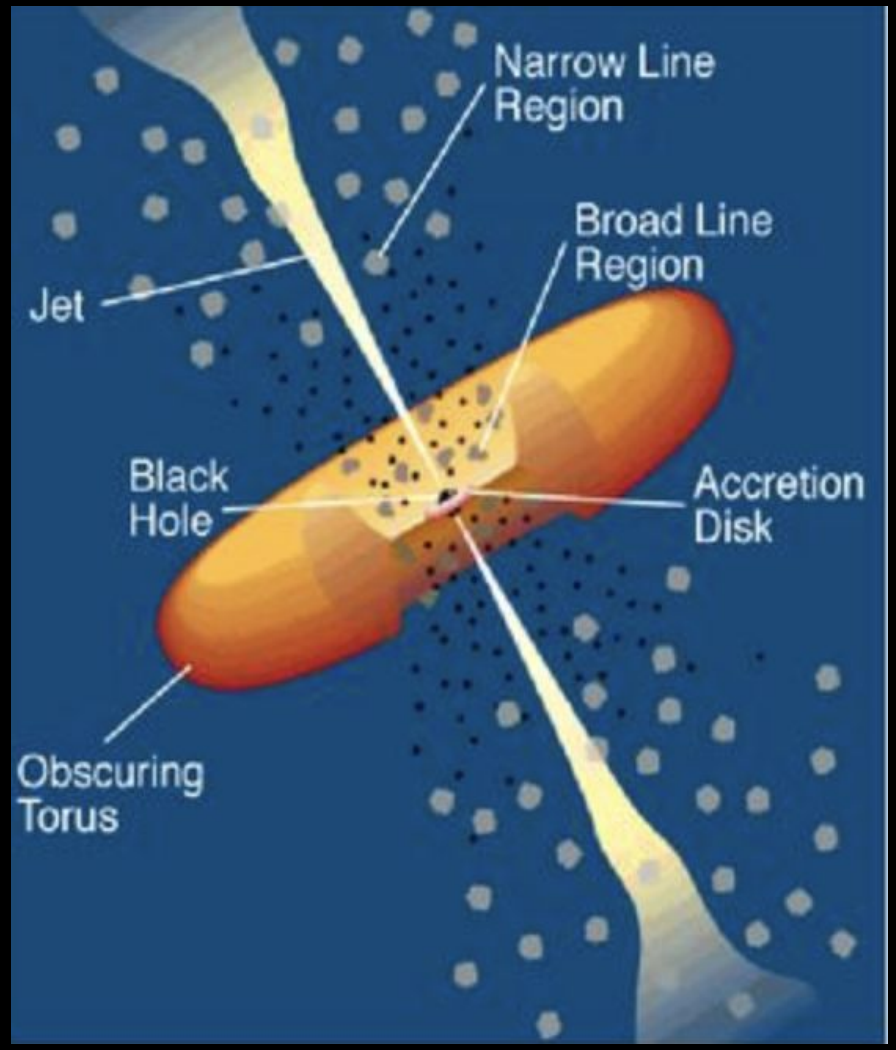
Masers

- Regions of intense, coherent radiation due to emission during an energy level transition.
- Good source for detecting protostellar activity.
- Emit at high brightness temperatures - great for detection.
- Properties of masers depend on the physical conditions of the emitting region - T, B, density.



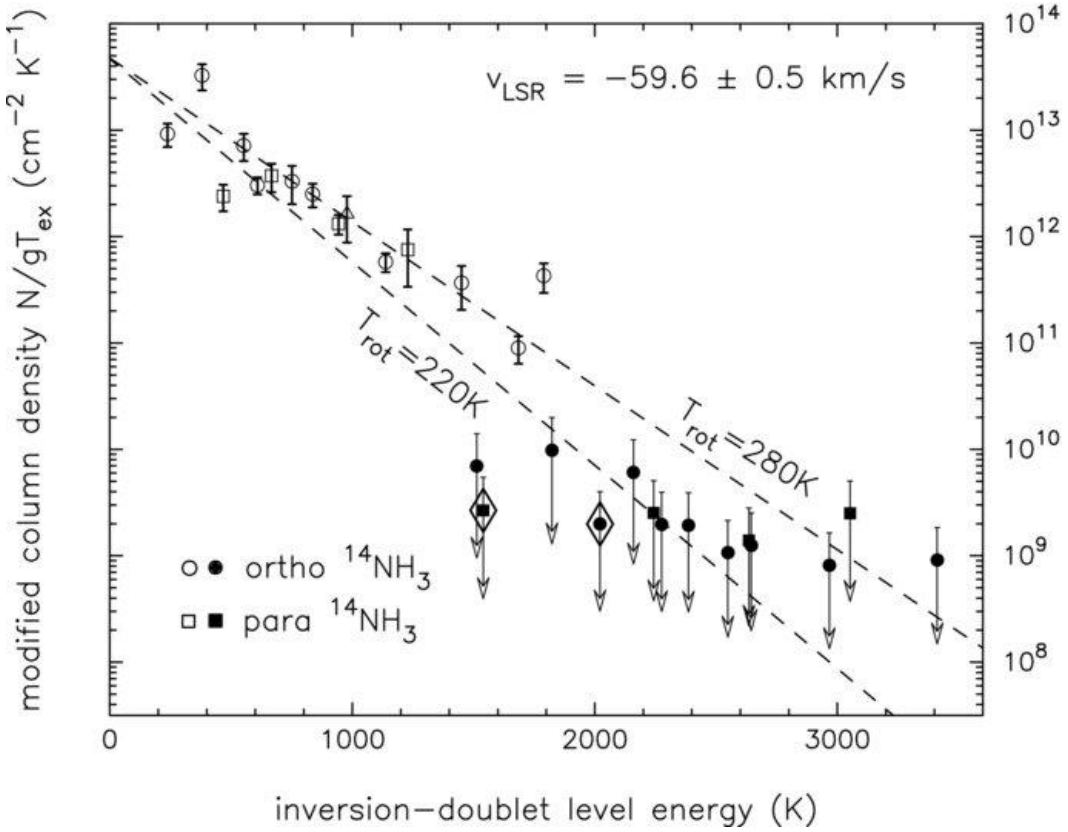
Credit: Taken from “A Search for water masers in High-redshift un-beamed AGNs”: T. Ghosh, S

Ammonia as a probe

- Ammonia is a great probe for massive star formation.
- Many spectral lines available in our frequency detection range.
- Non-metastable ammonia transitions are rare - 12 regions to date.

The (9,6) transition

- (9,6) is a non-metastable transition. They also seem to be the most intense ammonia lines in the W51-IRS2 region (Wootten et al. *(in review)*).
- Non-metastable ammonia population levels are dominated by collisional activity (Mauersberger et al. 1985; Ceccarelli et al. 2002).
- Due to the nature of the bond of NH3, the (9,6) is ortho-NH3.
- The emission seems to be completely non-thermal.



Boltzmann plot of the GBT observations of nonmetastable ammonia in NGC 7538.

Credit: Hoffman, Ian & Joyce, Spenser. (2014). New Maser Emission from Nonmetastable Ammonia in NGC 7538. IV. Coincident Masers in Adjacent States of Para-ammonia. The Astrophysical Journal. 782. 10.1088/0004-637X/782/2/83.

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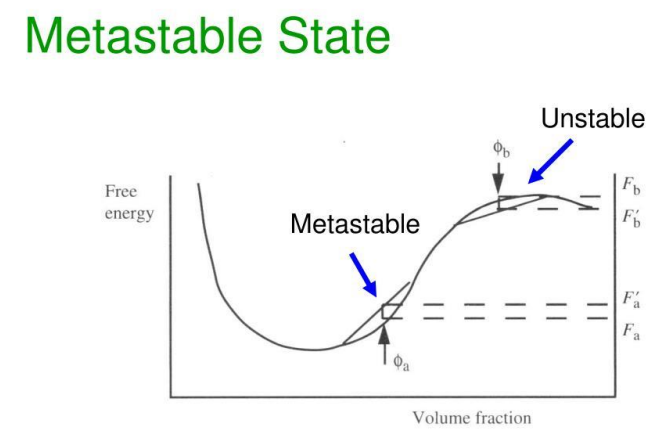
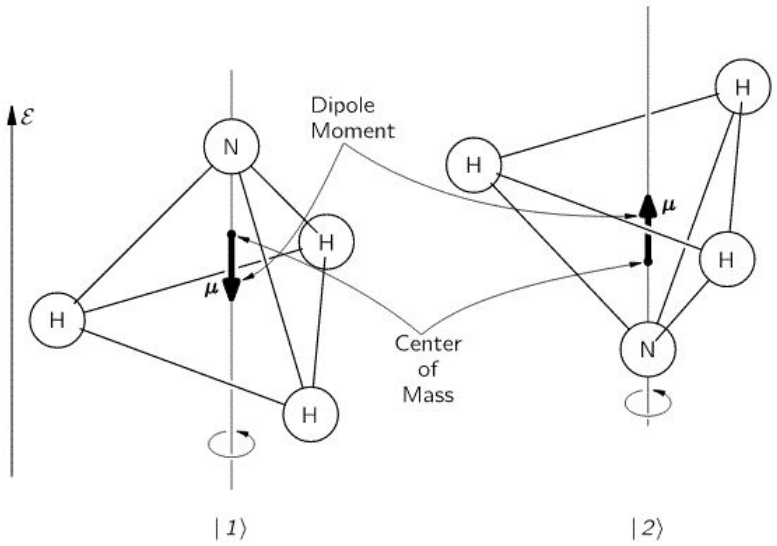
Transitions

- Ammonia energy level transitions take place due to many factors - incident radiation, chemical bonding, quantum tunneling etc.
- Metastable transition: When the total angular momentum quantum number, J is equal its projection on the molecular axis, K. [Metastable -> J=K]
- Non-metastable: J > K or J ≠ K. These transitions are unstable and eventually decay until J = K.



Credit: Taken from “Phase Transitions”, Department of Physics, Tunghai University



Credit: Taken from “The ammonia maser: transitions in a time-dependent field”, Jean Louis Van Belle

Motivation

- This work is a part of a larger survey project - **VLA 24A-399: *Flares, Shocks and YSOs: a systematic VLA survey of Ammonia Masers*** (Svoboda et al.).
- 28 targets, 60 lines (NH3, H2O, CH3OH). Sample selected from NOEMA+30m CORE (Beuther et al. 2018), ALMA-IMF Survey (Motte et al. 2022)
- Luminous targets: 10^4 M_{sol} - 10^6 M_{sol}

Research Goals

- Focusing on the (9,6) non-metastable transition for W51- IRS2.
- Derive maser properties & compare with works from previous epochs to document new masers, proper motion, and get better astrometric information.

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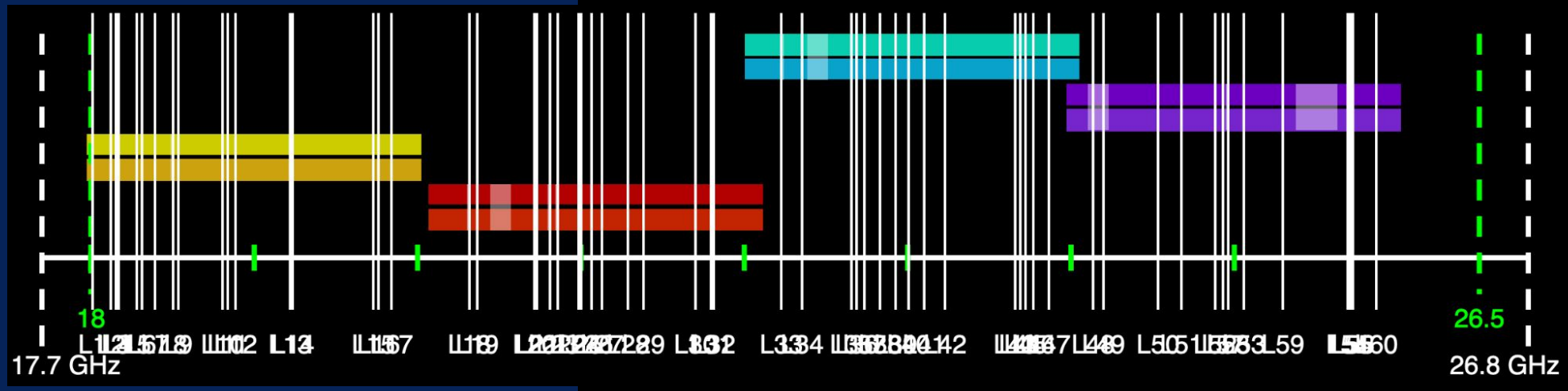
Observations



- 28 targets, 60 lines, 24 SBs
- VLA B-Config K-Band (18-26 GHz)
- 0.28” resolution
- Sample selected from NOEMA+30m CORE (Beuther et al. 2018), ALMA-IMF Survey (Motte et al. 2022)
- 5 mJy sensitivity
- At least 30 mins on source

- Worked on building a queryable database for known calibrators for future survey-like configurations.

calibrators

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3	ObjectId('66636a06b804b04…	"2050+314"	"J2000"	"A"	"20h50m51.131466s"	"31d27'27.372780""	"Aug01"
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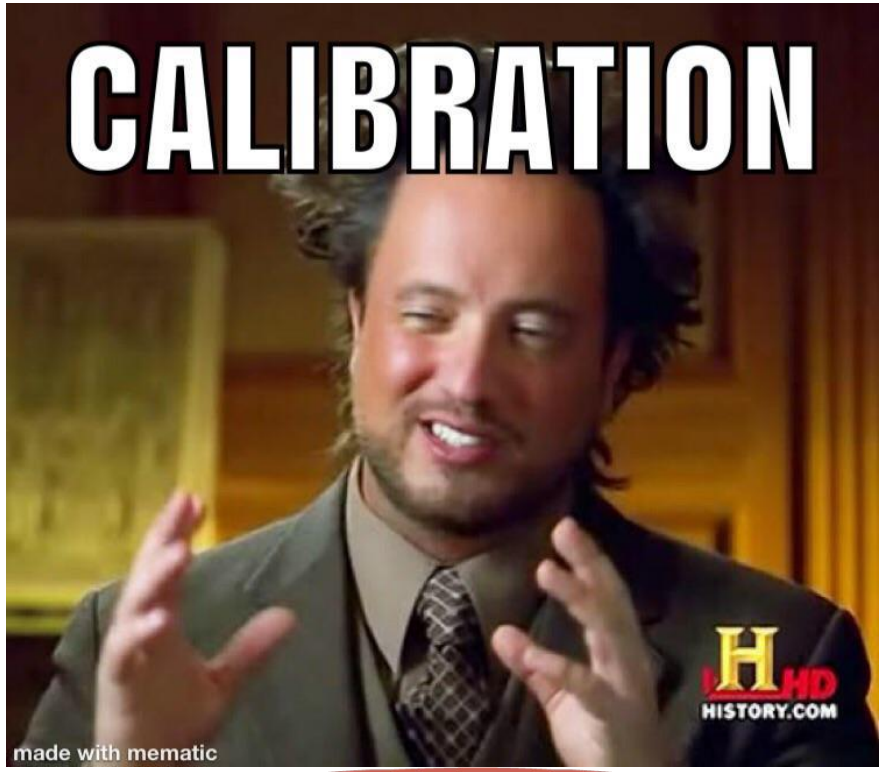
Calibration & Self-Calibration



- SRDP pipeline - Flagged out the ~2000 Jy H2O maser due to RFI.
- Modified NRAO pipeline for spectral analysis - flagged 1/2 - ¾ of basebands completely.
- Manual calibration: 2 execution blocks, (9,6) spw only.

Self-Calibration

- Greatest time consumption.
- Self-calibrations on a single channel would not apply to the rest of the spectral window.
- Workaround to apply self-cal solution.
- Poor noise still.

Results

W51-IRS2

A 300 x 300 pix cutout of the (9,6) masers in the central W51-IRS2 region at a frequency of ~18.5 GHz.

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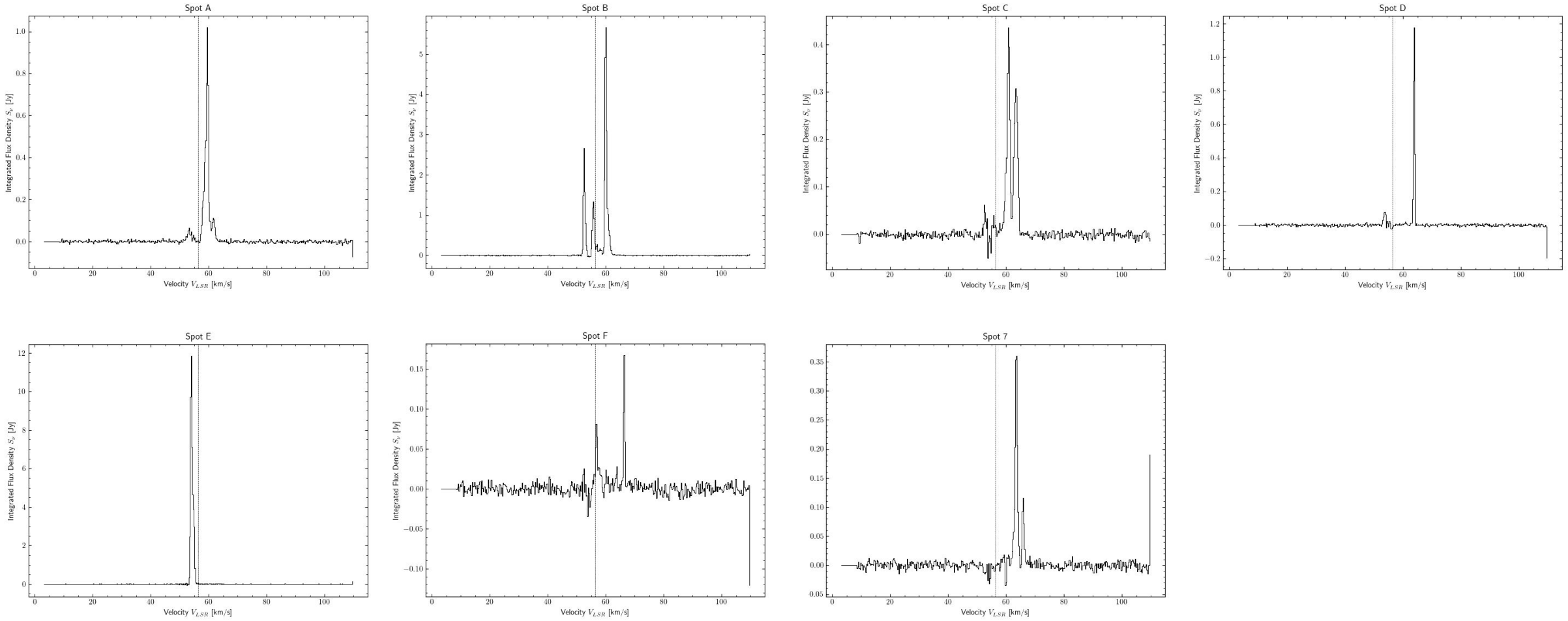

Spectra of detected masers in the central region, organized E-W. 7 spots detected in the central region. Scales vary between 0.25 Jy/beam - 12 Jy/beam in Intensity. Line widths 0.7 km/s to 3 km/s.

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Table of Observed Maser spots in the central 300 pix region for W51-IRS2

Maser Spot	Peak Pos (RA J2000)	Peak Pos (Dec J2000)	I_ν (Jy/beam)	T_{MB} (K)	V_{LSR}^{peak} (km s^{-1})	$\Delta V_{1/2}$ (km s^{-1})
A	19:23:40.076(0.0033 s)	14:31:05.336(0.050)"	1.020(0.001)	3208	59.42(0.01)	3.04(0.001)
B	19:23:40.037(0.0026 s)	14:31:05.716(0.042)"	5.995(0.001)	18859	59.92(0.01)	2.28(0.001)
C	19:23:39.957(0.0022 s)	14:31:05.323(0.036)"	0.471(0.001)	1481	60.68(0.01)	2.03(0.001)
D	19:23:39.832(0.0018 s)	14:31:05.300(0.029)"	1.274(0.001)	4008	63.72(0.01)	1.01(0.001)
E	19:23:39.815(0.0018 s)	14:31:04.810(0.031)"	12.353(0.001)	38861	53.85(0.01)	2.03(0.001)
F	19:23:39.744(0.0018 s)	14:31:05.156(0.026)"	0.18(0.001)	566	66.25(0.01)	0.76(0.001)
7	19:23:39.720(0.0020 s)	14:31:05.170(0.031)"	0.364(0.001)	1145	63.47(0.01)	1.77(0.001)

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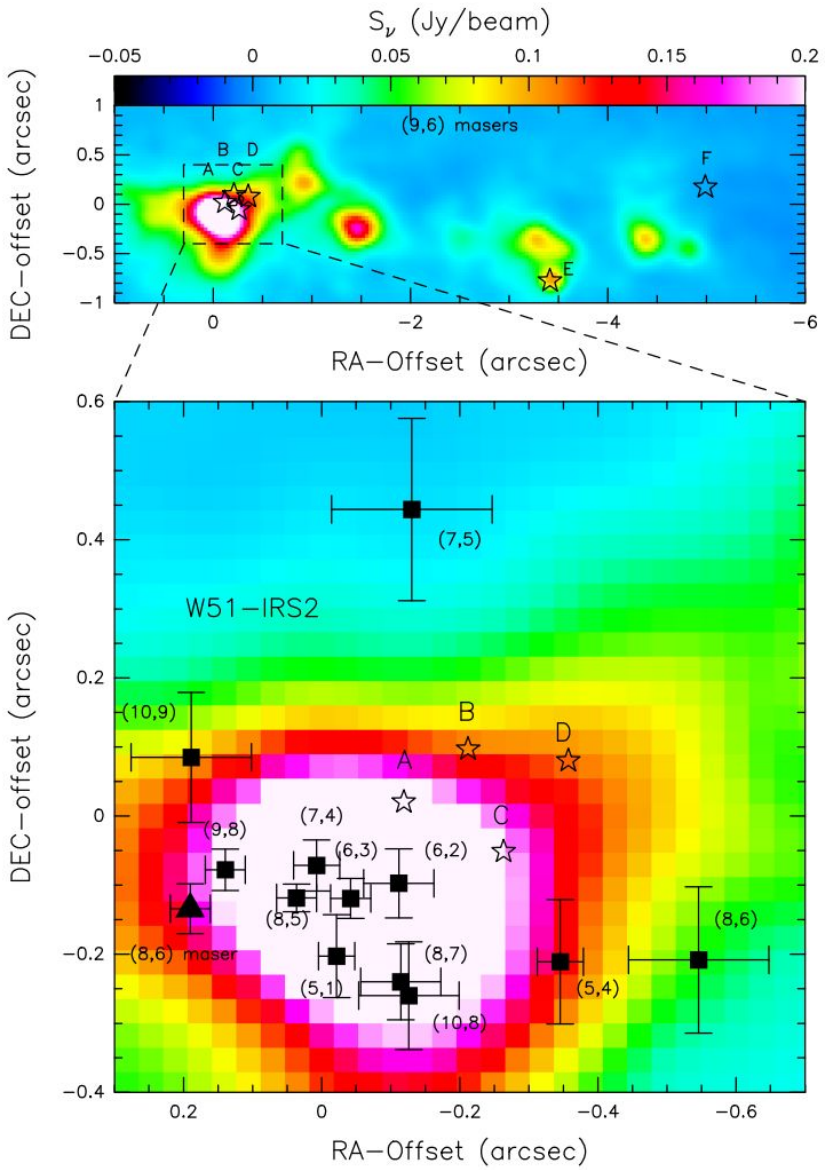

Fig. 2. *Upper panel:* The positions of the (9,6) line masers in W51-IRS2, superposed on the $\lambda \sim 1.3$ mm continuum map of Ginsburg et al. (2017). Offsets are relative to $\alpha_{J2000} = 19^h23^m40^s.055$. $\delta_{J2000} = 14°31'05''.59$, the SiO maser position from Eisner et al. (2002). For features A, C, D and E, where two measurements have been taken (Table 3), peak flux density weighted average positions are presented. **With the exception of feature D (see Fig. 4)** positional errors are too small to be shown on this scale. *Lower panel:* A zoom into the region with (9,6) masers A, B, C, and D, superposed on the 1.3 mm continuum (Ginsburg et al. 2017), also indicating the positions of the NH_3 quasi-thermal maxima. For masers A, C, and D, again averaged coordinates are given. For feature B with only one measurement (Table 3), uncertainties are about as large as the size of the symbol. Data for the quasi-thermal lines (filled squares) are given in Table 2, labelled 'th'. Error bars represent 2σ values deduced from Gaussian fits. We also include the position of the (8,6) maser line at about ($0''.2$, $-0''.1$) as a filled black triangle with 2σ error bars, while the (10,9) maser line at ($0''.9$, $-0''.1$) is, **in contrast to the line's quasi-thermal peak, well** outside of the shown window. Our peak of the continuum source W51d (see Fig. 1) relative to the origin of this plot is ($\Delta\alpha$, $\Delta\delta$) = ($-2''.25$, $2''.91$).

Comparison with Wootten et al. (in review)

Map made using 2013 data, VLA C-config, K-Band, overlaid on ALMA 230 GHz continuum.

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Maser Spot	Peak Pos (RA J2000)	(Dec J2000)	I_ν (Jy/beam)	T_{MB} (K)	V_{LSR}^{peak} (km s^{-1})	$\Delta V_{1/2}$ (km s^{-1})
A	19:23:40.076(0.0033 s)	14:31:05.336(0.050)"	1.020(0.001)	3208	59.42(0.01)	3.04(0.001)
B	19:23:40.037(0.0026 s)	14:31:05.716(0.042)"	5.995(0.001)	18859	59.92(0.01)	2.28(0.001)
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7	19:23:39.720(0.0020 s)	14:31:05.170(0.031)"	0.364(0.001)	1145	63.47(0.01)	1.77(0.001)

Comparison with Wootten et al. (in review)

Table of maser properties across 2 epochs of observation (2013, 2014).

(1) (J,K)[a]	(2) Date year/month/day	(3) ortho or para	(4) R.A.[b] J2000.0 19^h 23^m	(5) Dec.[b] J2000.0 $+14^o$ 31^s	(6) $I_\nu^{(c)}$ (Jy/beam)	(7) $T_{MB}^{(d)}$ (K)	(8) $V_{LSR}^{(c)}$ (km s^{-1})	(9) $\Delta V_{1/2}^{(c)}$ (km s^{-1})
(9,6)A	13/06/15	o	40.0504(0.002)^s	05.6603(0.024)$''$	0.9(0.01)	2800	59.5(0.004)	0.9(0.01)
(9,6)A	14/11/09	o	40.0432(0.002)	05.5611(0.030)	2.2(0.01)	6800	60.3(0.004)	1.3(0.01)
(9,6)B	13/06/15	o	40.0404(0.0013)	05.6873(0.021)	1.0(0.01)	3100	60.84(0.004)	0.8(0.01)
(9,6)C	13/06/15	o	40.0361(0.001)	05.5939(0.016)	1.3(0.01)	4100	55.7(0.004)	1.4(0.01)
(9,6)C	14/11/09	o	40.0376(0.001)	05.58378(0.007)	2.3(0.01)	7100	55.4(0.004)	1.1(0.01)
(9,6)D	13/06/15	o	40.0224(0.001)	05.8793(0.015)	1.4(0.01)	4300	51.53(0.004)	1.3(0.01)
(9,6)D	14/11/09	o	40.0384(0.001)	05.4616(0.01)	0.8(0.01)	2500	51.5(0.004)	3.7(0.01)
(9,6)E	13/06/15	o	39.8192(0.0003)	04.9021(0.005)	4.0(0.01)	12500	56.4(0.002)	0.8(0.01)
(9,6)E	14/11/09	o	39.8205(0.0006)	04.7299(0.005)	0.3(0.01)	900	53.7(0.002)	1.0(0.01)
(9,6)F	13/06/15	o	39.7114(0.004)	05.7662(0.064)	0.2(0.01)	600	62.5(0.010)	0.7(0.02)

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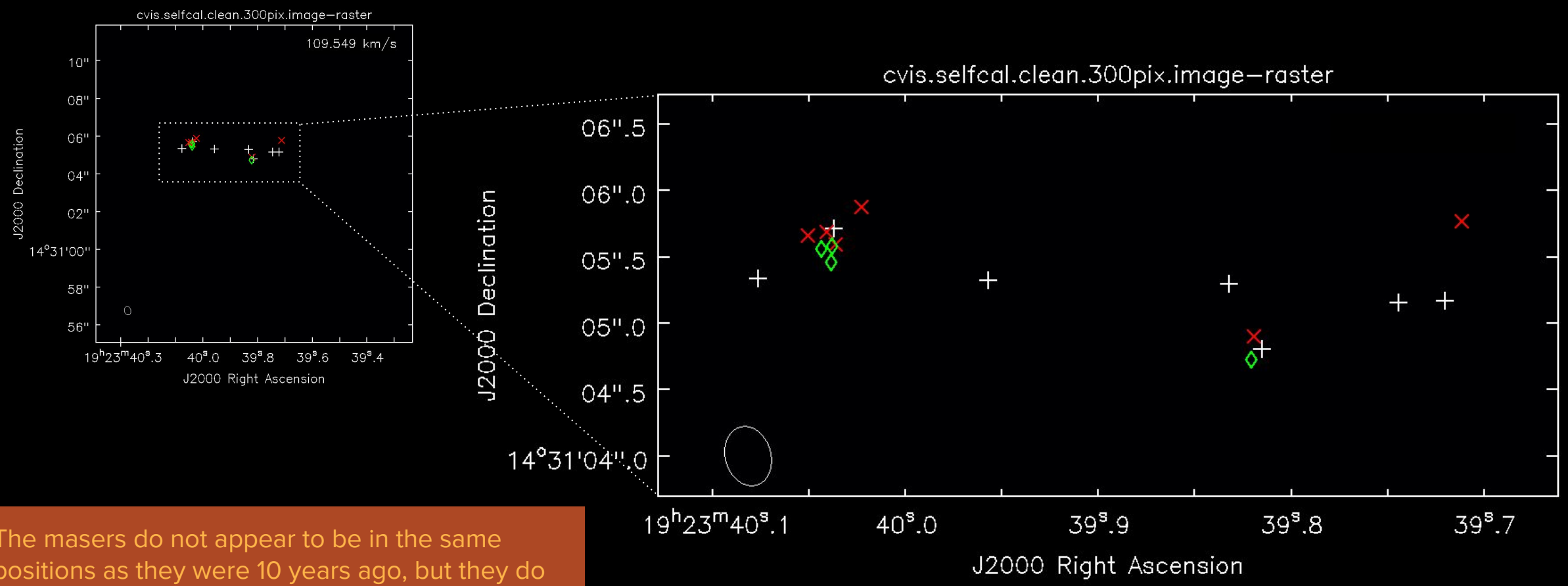

The masers do not appear to be in the same positions as they were 10 years ago, but they do not appear to move at consistent rates. The directions that they move in with respect to each other also does not currently provide us with information for the cause of this proper motion.

White Plus: Our observations (2024);
Red Cross: Wootten et al. (2013);
Green Diamond: Wootten et al (2014)

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Discussion of Results

Maser Spot	Δ RA (mas)		Δ Dec (mas)		Δ RA per Year (mas/year)		Δ Dec per Year (mas/year)	
	(2013 - 2024)	(2014 - 2024)	(2013 - 2024)	(2014 - 2024)	(2013 - 2024)	(2014 - 2024)	(2013 - 2024)	(2014 - 2024)
A	384.0	492.0	-324.3	-225.1	34.91	49.20	-29.48	-22.51
B	-51.0	-93.0	28.7	154.9	-4.64	-9.30	2.61	15.49
C	-1186.5	-1186.5	-270.9	-270.9	-107.86	-118.65	-24.63	-27.09
D	-2994.0	-3084.0	-579.3	-237.8	-272.18	-308.40	-52.66	-23.78
E	-253.5	46.5	80.1	80.1	-23.05	4.65	7.28	8.01
F	489.0	489.0	-610.2	-610.2	44.45	48.90	-55.47	-61.02

- There is a range of proper motion, i.e., not all masers are moving away at the same rate.
- Lowest proper motion detected at the center. (BLUE)

- Highest proper motion detected in spots closest to the center. (RED) OR possible maser has dimmed and this is a new detection.
- 3 mas/yr - 300 mas/yr in a 84” central cutout. 16.5 AU/yr - 1650 AU/yr. 78.21 km/s - 7821 km/s

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Conclusions

- Our detections presented show highly point-source objects, with sharp and narrow spectra, and extremely high temperatures - all characteristics of non-thermal emission.
- Our higher resolution, higher sensitivity data has found more (9,6) masers.
- The presence of the 12 Jy maser indicates high star formation activity, and along with detections of the H2O maser, will inform our research of kinematics in the W51 star-forming region.

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Future Work

- Mapping the continuum to check for possible driver of proper motion.
- Image the H2O masers in this region (~2000 Jy).
- Image the other lines from our observations and check for quasi-thermal lines and other phenomena in this region.
- Perform a comparison for other lines that are common with the Wootten et al. observations of W51.
- Process the data of the other 27 targets in our survey.

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Acknowledgements

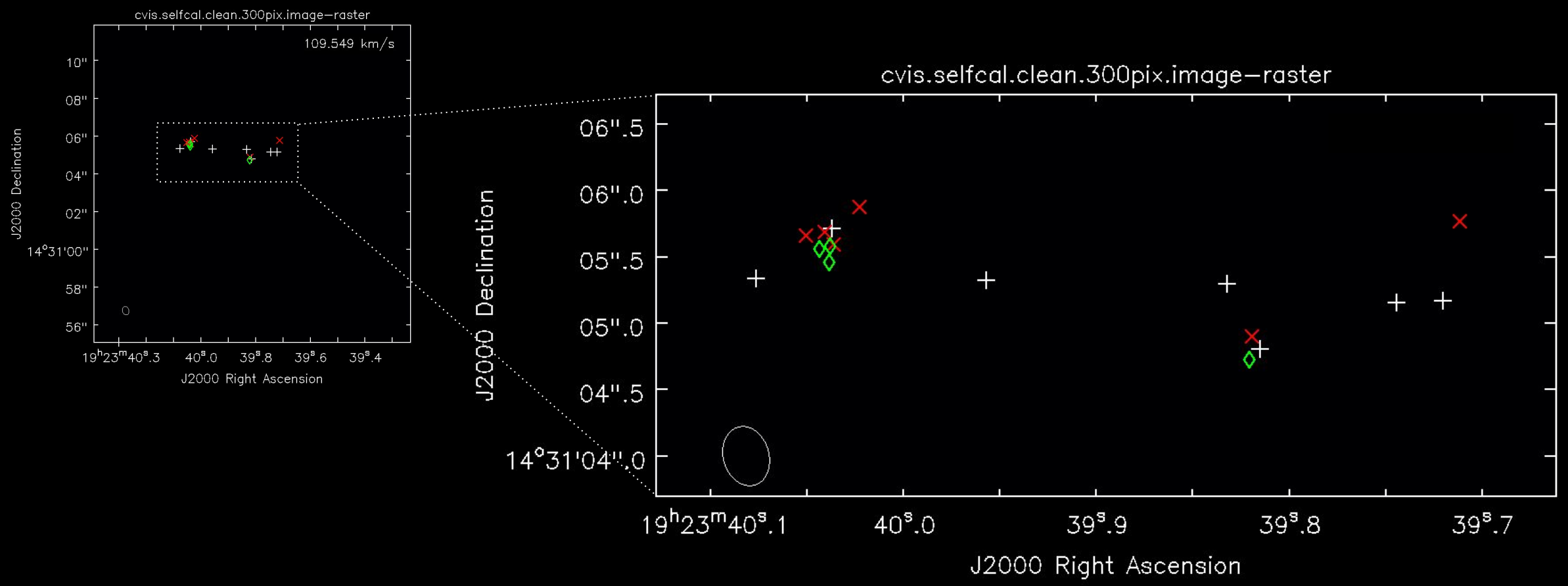
I would like to thank the NSF and AUI for providing me this summer scholarship as an international graduate student. I would like to thank the summer students and the staff at the NRAO for allowing me to ask questions when I needed help.

I would like to thank Anna Kapinska for helping with the administrative difficulties in order to sign up for the research program.

I would like to thank my Department Advisor, Peter Hofner for encouraging me to apply for this program and support my research endeavors.

I would like to thank Al Wootten for being so cooperative and allowing us access to their research for advisement purposes.

I would like to thank my NRAO advisors Brian Svoboda and Emmanuel Momjian, who have gone out of their way to help me with this research project, guide me, and potentially build it into a career-defining track for me.


cvis.selfcal.clean.300pix.image−raster
109.549 km/s
10''
08''
06''
04''
02''
14°31'00''
58''
56''
J2000 Declination
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40s.0
39s.8
39s.6
39s.4
J2000 Right Ascension
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06''.5
06''.0
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05''.0
04''.5
14°31'04''.0
J2000 Declination
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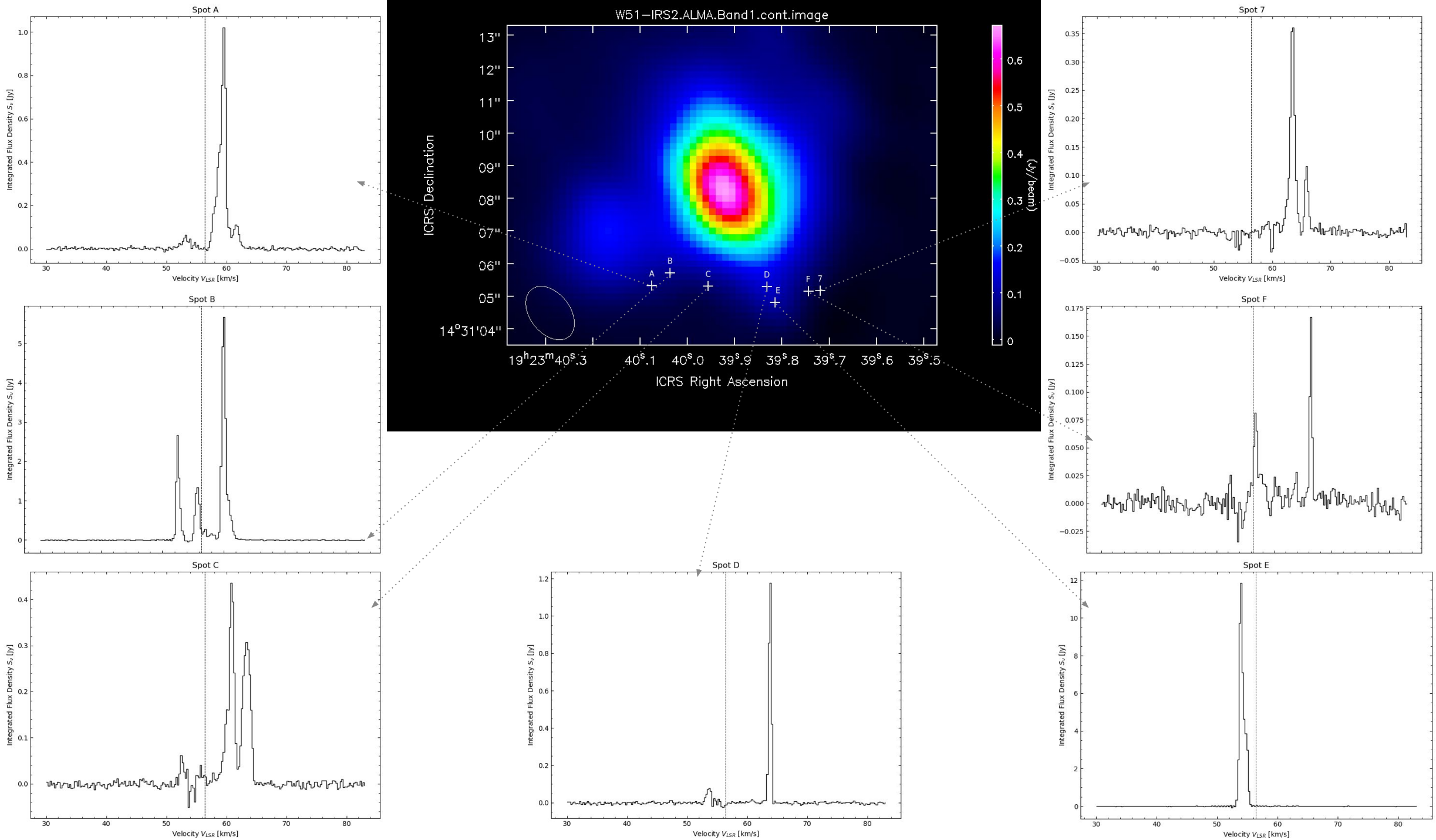
40s.0
39s.9
39s.8
39s.7
J2000 Right Ascension

(1) (J,K)[a]	(2) Date year/month/day	(3) ortho or para	(4) R.A.[b] J2000.0 $19^h\ 23^m$	(5) Dec.[b] J2000.0 $+14^o\ 31^s$	(6) $I_\nu^{(c)}$ (Jy/beam)	(7) $T_{MB}^{(d)}$ (K)	(8) $V_{LSR}^{(c)}$ (km s^{-1})	(9) $\Delta V_{1/2}^{(c)}$ (km s^{-1})
(9,6)A	13/06/15	o	40.0504(0.002)^s	05.6603(0.024)$''$	0.9(0.01)	2800	59.5(0.004)	0.9(0.01)
(9,6)A	14/11/09	o	40.0432(0.002)	05.5611(0.030)	2.2(0.01)	6800	60.3(0.004)	1.3(0.01)
(9,6)B	13/06/15	o	40.0404(0.0013)	05.6873(0.021)	1.0(0.01)	3100	60.84(0.004)	0.8(0.01)
(9,6)C	13/06/15	o	40.0361(0.001)	05.5939(0.016)	1.3(0.01)	4100	55.7(0.004)	1.4(0.01)
(9,6)C	14/11/09	o	40.0376(0.001)	05.58378(0.007)	2.3(0.01)	7100	55.4(0.004)	1.1(0.01)
(9,6)D	13/06/15	o	40.0224(0.001)	05.8793(0.015)	1.4(0.01)	4300	51.53(0.004)	1.3(0.01)
(9,6)D	14/11/09	o	40.0384(0.001)	05.4616(0.01)	0.8(0.01)	2500	51.5(0.004)	3.7(0.01)
(9,6)E	13/06/15	o	39.8192(0.0003)	04.9021(0.005)	4.0(0.01)	12500	56.4(0.002)	0.8(0.01)
(9,6)E	14/11/09	o	39.8205(0.0006)	04.7299(0.005)	0.3(0.01)	900	53.7(0.002)	1.0(0.01)
(9,6)F	13/06/15	o	39.7114(0.004)	05.7662(0.064)	0.2(0.01)	600	62.5(0.010)	0.7(0.02)


W51−IRS2.ALMA.Band1.cont.image
ICRS Declination
ICRS Right Ascension
(Jy/beam)
A
B
C
D
E
F
7
Spot A
Spot B
Spot C
Spot D
Spot E
Spot F
Spot 7
Integrated Flux Density S_ν [Jy]
Velocity V_{LSR} [km/s]